

Def. A map

$$\exp: \mathbb{C} \rightarrow \mathbb{C}: z \mapsto \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

is called the exponential, observe the following: given a real-value x ,

$$\begin{aligned} e^{ix} &= \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n}}{(2n)!} + i \cdot \left(\sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!} \right) \\ &= \cos x + i \cdot \sin x \end{aligned}$$